

Same FIN Heat Flow, Different Fluctuation Actions

Cosh dissipation, integer-count constraints,
and exact coarse current contraction

Programme ST8548

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

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An analytical and computational follow-up to ST8547. The report tests whether the quadratic logarithmic-mean transport cost is a microscopic fluctuation law. It derives an alternative from independent Markov walkers, identifies observable differences, and proves exact current contraction through product refinement. The microscopic realization remains a supplied model.

Abstract

The logarithmic mean gives a valid quadratic Onsager representation of finite graph heat flow. Independent continuous-time Markov walkers with the same conductances instead yield a nonquadratic fluctuation action with a cosh dissipation potential. Both have the same mean current and the same time-reversal cost asymmetry. Their local variances differ by $z \coth z \geq 1$, where z is half the log population ratio. For the explicit strict FIN test the ratio is 1.4683539294. The quadratic prescription even violates the variance lower bound for integer-valued Markov counts when $|\log(a/b)| > 2$.

The exact jump action contracts under summing fine channels for arbitrary positive hidden conditional distributions. Its covariance also projects exactly. This differs from the quadratic Onsager equality criterion in ST8547, which requires independence. Thus a defect of one gradient metric does not imply failure of exact stochastic coarse-graining.

All conclusions are conditional on supplied rates and a microscopic counting model. The underlying nonquadratic gradient structure is established literature. This study provides a checked application, a counting test and corrections to FIN's interpretation; it makes no claim of new fundamental physics or mathematical priority.

Contents

1	Scope, source evidence and conventions	2
2	Exact local current fluctuations	2
3	The nonquadratic entropy-dissipation structure	3
4	Quadratic mobility does not fix microscopic noise	3
4.1	Whole-graph covariance	4
5	Exact coarse current contraction	4
6	Strict numerical example and independent checks	5
7	A necessary correction to the older kinetic example	6
8	Remaining assumptions and the next physical test	6
A	Reproduction	7

1 Scope, source evidence and conventions

The project state map in AGENTS.md, the ST8547 proofs and code, and the earlier configuration-generator examples were consulted. This is a focused proof-level continuation, not a new verification of every historical FIN report. The raw archived status labels and hashes are not treated as independent proof certificates.

The basic microscopic/gradient distinction was established by Mielke, Peletier and Renger [2], with further developments by Liero et al. [3]. Maas [1] supplies the logarithmic-mean quadratic representation. The present derivations and numerical tests state exactly which of these structures is being used.

The strict conductances are

$$W_{ij} = \frac{\cos(0.18575d + 0.16250)}{1 + d^{9/5}}, \quad d = \min(|i - j|, 12 - |i - j|), \quad W_{ii} = 0.$$

They are positive on $d = 1, \dots, 6$. Let $Q = W - \text{diag}(W\mathbf{1}) = -A$, a row-generator convention. The raw legacy kernel has signed weights and is not substituted into this positive-rate construction.

Assume N independent Markov walkers with generator Q . Their empirical occupation vector has increments $(e_j - e_i)/N$, occurring at rate $Np_i W_{ij}$. Its exact generator is

$$(\mathcal{G}_N F)(p) = N \sum_{i \neq j} p_i W_{ij} \left[F\left(p + \frac{e_j - e_i}{N}\right) - F(p) \right]. \quad (1)$$

Coordinate functions give the mean equation $\dot{p} = -Ap$. The population, independence assumption and rate interpretation of W are additional model premises; physical time is not derived.

2 Exact local current fluctuations

Orient edge $i \rightarrow j$. Write

$$a = W_{ij}p_i, \quad b = W_{ij}p_j, \quad a, b > 0.$$

The local current Hamiltonian is

$$h(s; a, b) = a(e^s - 1) + b(e^{-s} - 1). \quad (2)$$

It follows either from the two oriented generator increments or from the nonlinear exponential transform of (1). Derivatives at zero give instantaneous cumulant rates

$$\kappa_1 = a - b, \quad \kappa_2 = a + b, \quad \kappa_3 = a - b, \quad \kappa_4 = a + b.$$

For N independent walkers the total-count cumulants scale by N .

This is an exact *local* action density. For a closed chain it is not the finite-time count distribution at arbitrary time: the state changes and return transitions occur. Section 6 checks finite-time counts with the full tilted generator.

Theorem 1 (Current rate function). *The Legendre transform of (2) is*

$$\ell(j; a, b) = j \operatorname{arsinh} \frac{j}{2\sqrt{ab}} - \frac{j}{2} \log(a/b) - \sqrt{j^2 + 4ab} + a + b. \quad (3)$$

It is strictly convex in j , with unique zero $j_0 = a - b$.

Proof. The maximizing tilt solves $j = ae^s - be^{-s}$, so

$$s_* = \operatorname{arsinh} \frac{j}{2\sqrt{ab}} - \frac{1}{2} \log(a/b).$$

At that tilt $ae^s + be^{-s} = \sqrt{j^2 + 4ab}$, yielding (3). Its second derivative is $1/\sqrt{j^2 + 4ab} > 0$, and direct substitution at $a - b$ gives zero. $\square$

There is an independent flux representation:

$$\ell(j; a, b) = \inf_{u, v \geq 0: u-v=j} [u \log(u/a) - u + a + v \log(v/b) - v + b].$$

The minimizers are $u = (\sqrt{j^2 + 4ab} + j)/2$, $v = (\sqrt{j^2 + 4ab} - j)/2$. Relative-entropy perspective convexity proves joint convexity in (j, a, b) . Also $\ell(cj; ca, cb) = c\ell(j; a, b)$.

3 The nonquadratic entropy-dissipation structure

Set $g = 2\sqrt{ab}$, $z = \frac{1}{2} \log(a/b)$, and define

$$\mathcal{R}(j) = j \operatorname{arsinh}(j/g) - \sqrt{j^2 + g^2} + g, \quad \mathcal{R}^*(\zeta) = g(\cosh \zeta - 1).$$

These are symmetric Legendre-dual potentials, and

$$\ell(j; a, b) = \mathcal{R}(j) + \mathcal{R}^*(z) - jz, \quad j_0 = \partial_\zeta \mathcal{R}^*(z) = a - b.$$

The force z corresponds to one-half of negative relative Shannon entropy in the energy-dissipation convention. This factor follows from the reversal identity; it is not a physical constant.

The same microscopic independence premise also gives relative entropy from equilibrium counting. For invariant law π , independent occupations are multinomial:

$$\Pr(Np) = \frac{N!}{\prod_i (Np_i)!} \prod_i \pi_i^{Np_i}.$$

Stirling's formula yields $-N^{-1} \log \Pr(Np) \rightarrow D(p||\pi)$. Here $\pi_i = 1/12$ since W is symmetric. Thus a microscopic counting assumption can select a conditional entropy and dissipation structure together. It is not furnished by A alone.

4 Quadratic mobility does not fix microscopic noise

Let $m = \Lambda(a, b) = (a - b)/\log(a/b)$, with diagonal value $m = a$. To compare fairly, normalize the quadratic action as

$$\ell_Q(j; a, b) = \frac{[j - (a - b)]^2}{4m}. \quad (4)$$

It has the same mean and exactly the same reversal asymmetry:

$$\ell(j) - \ell(-j) = \ell_Q(j) - \ell_Q(-j) = -j \log(a/b).$$

The factor 4 compensates the one-half entropy convention; it is chosen explicitly rather than hidden in a clock.

Theorem 2 (Variance mismatch). *The jump action and the quadratic prescription have local variances*

$$v_{\text{jump}} = a + b, \quad v_Q = 2\Lambda(a, b).$$

Their ratio is

$$\boxed{\frac{v_{\text{jump}}}{v_Q} = z \coth z \geq 1,}$$

with equality precisely at $a = b$.

Proof. Inverse rate-function curvature at j_0 equals $a+b$ for (3) and $2m$ for (4). Write $a = \sqrt{abe^z}$, $b = \sqrt{abe^{-z}}$. The ratio becomes $z/\tanh z$, and $|\tanh z| \leq |z|$, strictly off zero. $\square$

Near equilibrium this ratio is $1 + z^2/3 + O(z^4)$. Even at equilibrium, the exact fourth cumulant rate is $2a > 0$, while the local quadratic Hamiltonian has zero fourth cumulant. The deterministic ST8547 gradient identity remains valid.

Theorem 3 (Integer-count consistency). *For any Markov additive count with integer increments k , nonnegative local rates c_k , and finite first two moments,*

$$v = \sum_k k^2 c_k \geq \left| \sum_k k c_k \right| = |\mu|.$$

The quadratic prescription violates this necessary condition whenever

$$|\log(a/b)| > 2.$$

Proof. Use $k^2 \geq |k|$ and the triangle inequality. For (4), $v_Q/|j_0| = 2/|\log(a/b)|$, which is less than one in the stated regime. $\square$

This excludes an exact integer-valued Markov count interpretation in that regime, even allowing collective integer jumps. It does not exclude a continuous Gaussian approximation or another coarse observable.

Three means have distinct roles: the logarithmic mean relates mean current to finite entropy force; the geometric mean sets the cosh prefactor; the arithmetic combination determines local shot noise.

4.1 Whole-graph covariance

The empirical-occupation covariance rate per particle is

$$\Sigma(p) = \sum_{i < j} W_{ij}(p_i + p_j)(e_i - e_j)(e_i - e_j)^\top.$$

The quadratic prescription uses $2K_\Lambda(p)$. Arithmetic-logarithmic mean ordering yields

$$\Sigma(p) - 2K_\Lambda(p) \succeq 0.$$

The difference is a graph Laplacian with nonnegative conductances. For connected support it is zero exactly at uniform p .

5 Exact coarse current contraction

Let one coarse edge aggregate fine channels with positive rates a_b, b_b . Set $a = \sum_b a_b, b = \sum_b b_b$.

Theorem 4 (Current-action contraction).

$$\inf_{\sum_b j_b = j} \sum_b \ell(j_b; a_b, b_b) = \ell(j; a, b).$$

No equality of forward and reverse conditional child profiles is required.

Proof. Choose s with $ae^s - be^{-s} = j$, and $j_b = a_b e^s - b_b e^{-s}$. All fine costs have derivative s at this feasible vector, so convexity proves optimality. Their Legendre values sum to $sj - \sum_b h(s; a_b, b_b) = sj - h(s; a, b)$. $\square$

In the supplied product FIN graph, $a_b = W_{ij}P_{ib}$ and $b_b = W_{ij}P_{jb}$. Consequently this contraction holds for arbitrary strictly positive correlated fine states. Its covariance also satisfies

$$C\Sigma_{\text{fine}}(P)C^\top = \Sigma_{\text{coarse}}(CP),$$

since the edge coefficients are linear in P and vertical increments vanish after projection.

For $A \otimes I + I \otimes B$, the base-coordinate Markov process is exactly lumpable for every initial joint state. This is a product generator, not the interacting Ising/prism generator considered in older stationary-RG studies. Their different lumpability properties are not contradictory.

The quadratic contraction instead gives

$$\inf_{\sum_{j_b=j}} \sum_b \ell_Q(j_b; a_b, b_b) = \frac{[j - (a - b)]^2}{4 \sum_b \Lambda(a_b, b_b)}.$$

As shown in ST8547, $\sum_b \Lambda(a_b, b_b) \leq \Lambda(a, b)$. For $j \neq j_0$, equality of quadratic costs requires matching forward and reverse normalized child profiles.

Thus the ST8547 correlation defect is a defect of that quadratic representation; it does not force a defect of exact Markov counting statistics. Its KL entropy-production identity remains valid.

6 Strict numerical example and independent checks

Use $p_i = (i + 1)/78$ and edge $0 \rightarrow 11$.

Quantity	Value
$W_{0,11}$	0.4699856726450201
a	0.006025457341602822
b	0.07230548809923387
Mean current $a - b$	-0.06628003075763104
Jump variance rate	0.0783309454408367
Quadratic variance rate	0.05334609311241991
Jump/quadratic variance	1.4683539294201824

The population ratio is 12. The local count ratios are

$$\frac{v_{\text{jump}}}{|j_0|} = \frac{13}{11} > 1, \quad \frac{v_Q}{|j_0|} = \frac{2}{\log 12} < 1.$$

For forward child proportions (0.8, 0.2), reverse proportions (0.25, 0.75) and current $j = 0.04$, the exact contraction error is 2.78×10^{-17} . The quadratic fine-contracted and coarse costs are 0.1178653956 and 0.1058694675. The graph covariance projection error is 1.11×10^{-16} ; the quadratic projection defect norm is 0.0079424221.

Finite-time counts were independently computed using derivatives of the full tilted 12-state generator. For one walker with initial distribution p :

Window t	variance/ t	fourth cumulant/ t
0.1	0.0745294243	0.0717795571
0.01	0.0779222577	0.0776108767
0.0001	0.0783268250	0.0783236669
0.00001	0.0783305334	0.0783302175

The limits are the local jump value 0.0783309454. No finite-time Poisson approximation was substituted for this calculation.

Eighteen tests independently check the Legendre transform by optimization, the flux-relative-entropy contraction, high-precision derivatives, the cosh decomposition, convexity, integer-count bound, covariance order, projection, finite-time tilted moments and the correction below. These are numerical checks supporting explicit analytical proofs, not interval certificates or physical evidence.

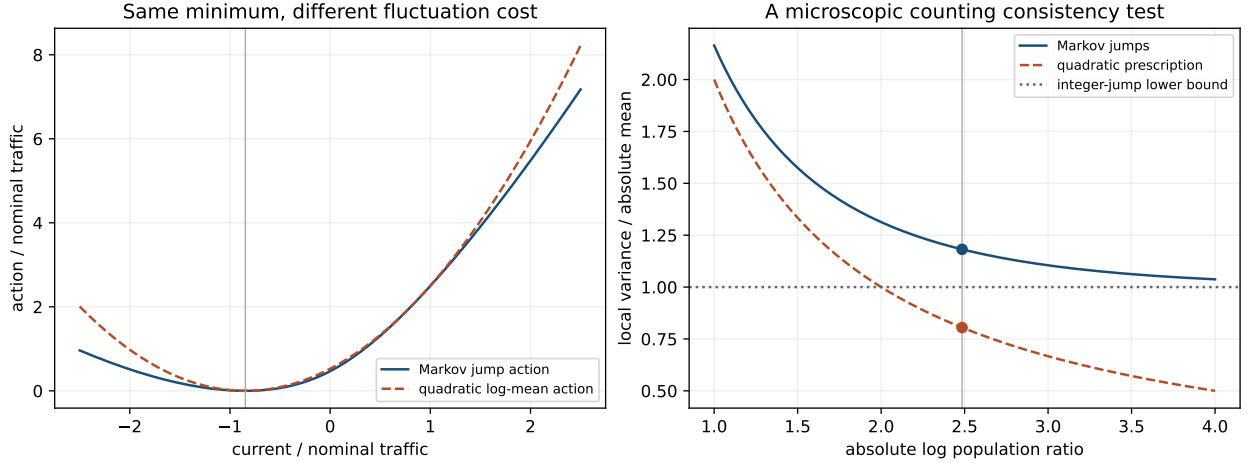


Figure 1: The exact and quadratic current costs share the deterministic minimum (left). Their local variance-to-mean ratios differ, and the quadratic prescription crosses below the integer-jump bound (right).

7 A necessary correction to the older kinetic example

In ST2547–ST2561, the three-bit model has energy $E = -\theta x_1 x_2 x_3$. Every flip has $|\Delta E| = 2|\theta|$. Therefore

$$Q_{\text{Met}} = (1 + e^{-2|\theta|})Q_{\text{HB}}, \quad Q_{\text{sym}} = 2 \cosh(\theta)Q_{\text{HB}}.$$

They have identical normalized spectral shapes. The earlier claim that this example distinguished more than a clock factor is false. At $\theta = 0.3$, both proportionality identities replay within 4.45×10^{-16} .

General kinetic nonuniqueness remains, but that specific example did not prove it beyond time scaling. Historical files are preserved and this correction supersedes their interpretation.

8 Remaining assumptions and the next physical test

Microscopic independence is essential. If particles instead move in synchronized groups of size s , the local Hamiltonian per total particle is

$$h_s(\xi) = \frac{1}{s} \left[a(e^{s\xi} - 1) + b(e^{-s\xi} - 1) \right].$$

Its mean remains $a - b$, but its variance is $s(a + b)$. Hence the same mean heat equation does not determine a unique population law. The 12-state walker model is also distinct from a 4096-state binary configuration model.

A dimensionless conditional experiment can compare short-window net-count variance to the absolute mean net count. Population size and the common clock factor cancel. At zero drift, compare local fourth and second cumulants instead. Repeated preparation, bidirectional event counts, finite-window extrapolation and detector-loss checks are needed; no such laboratory record is claimed here.

The main result for FIN is an observable distinction between gradient representations with identical deterministic dynamics, and exact coarse contraction of the microscopic current action. The best next investigation is a finite-noise counting protocol testing single versus collective events and the predicted cumulants. A sourced FIN microscopic law must specify what those events are.

The strict $U_t = e^{-itA}$ and $P_t = e^{-tA}$ remain distinct functional calculi of one operator. This study concerns the stochastic realization of the heat channel. It derives neither quantum measurement dynamics nor physical units, selector/QW-2191, legacy role transfer, continuum spacetime, Standard Model, gravity, L_{total} , or a Theory of Everything.

A Reproduction

```
python3 analysis.py --output replay_results.json
python3 -m unittest -v test_analysis.py
MPLCONFIGDIR=/tmp/fin-st8548-mpl python3 plot_results.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The package includes the full proofs in PROOFS_EN.md, input hashes and dependency versions in provenance.json, and checksums. No network access or large computation is required for replay.

References

- [1] J. Maas, *Gradient flows of the entropy for finite Markov chains* (2011). <https://arxiv.org/abs/1102.5238>.
- [2] A. Mielke, M. A. Peletier and D. R. M. Renger, *On the relation between gradient flows and the large-deviation principle, with applications to Markov chains and diffusion*. <https://arxiv.org/abs/1312.7591>. See Section 4.1 and Example 4.2.
- [3] M. Liero, A. Mielke, M. A. Peletier and D. R. M. Renger, *On microscopic origins of generalized gradient structures*, Discrete and Continuous Dynamical Systems–S 10 (2017), 1–35. <https://doi.org/10.3934/dcdss.2017001>.